

**Mathematics Syllabus for Three Major B. Sc. Programme from the
Academic year 2024-25
(Semester Scheme)**

Preamble:

The B.Sc. Mathematics course aims to provide students with a comprehensive foundation in mathematical principles, theories, and applications. This program is designed to develop critical thinking, analytical skills, and problem-solving abilities essential for various scientific and technical careers.

The Mathematics syllabus for B.Sc. (Two Major Programme) in use at present was introduced from the academic year 2021-22 as per NEP-2020 structure and guidelines given by the state government in 2021. Based on the directions and guidelines from the Higher Education Council of the Government of Karnataka (GO: ED 166 UNE 2023 Bangalore, Date: 08.05.2024), Mangalore University has issued new guidelines to launch the Three Major B.Sc. degree programme starting from the academic year 2024-25. Consequently, the revised and restructured syllabus for Mathematics as an optional subject in the B.Sc. (Three Major Programme) has been prepared according to the new regulations of the University, by modifying the earlier syllabus, including Lab components and introducing new text and reference books.

The following new syllabus for Mathematics as an optional subject in the B.Sc. (Three Major Programme) at Mangalore University has been framed by the Board of Studies in Mathematics for the UG programme. This syllabus will be implemented starting from the academic year 2024-25.

Aims and objectives of the restructured syllabus

- Equip students with a deep understanding of core mathematical concepts and methodologies.
- Improve the perspective of students on mathematics as per modern requirement and develop a spirit of inquiry and scientific temper in the student.
- Initiate students to enjoy mathematics, pose and solve meaningful problems, to use abstraction to perceive relationships and structure and to understand the basic structure of mathematics.
- Create a student-friendly learning environment by encouraging experimental, problem-solving, and discovery-based approaches to learning mathematics.
- To orient students towards relating mathematics applications and improve retention of mathematical concepts in the student.

V Semester

BSCMTCS501	Ring theory and Laplace Transforms	3 Credits	(36 Hours, 3 hours/ week)
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Course Objectives:

The course will help the students

1. To introduce the fundamental concepts of Ring theory, including binary operations, Integral Domains, and fields.
2. To explore deeper aspects of Ring theory such as Ideals, homomorphisms, and isomorphism theorems.
3. To find Laplace transform and inverse Laplace transform of various functions
4. Able to take more courses on wave equation, heat equation, and Laplace equation.
5. To Solve Ordinary Differential equation by Laplace Transforms

Course Outcomes (COs):

1. Students will understand and apply concepts of Ring operations, fields, Integral domain
 2. Students will be able to work with ideals, kernel, homomorphisms, Isomorphisms
 3. Students will be able to find the Laplace transform of some standard functions, Step functions, Periodic functions
 4. Students will be able to solve ordinary differential equations using Laplace Transforms
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UNIT I (12 Hours)

Ring Theory: Commutative Ring, Unit element, Properties, Zero divisors, Integral domains, Field, Division ring (Skew field), regular elements, Finite Integral domains, Theorems.

Ring Homomorphisms: Kernel of a ring homomorphism, Isomorphism Ideals, Simple Rings, Left and right ideals, Sum and Product of two ideals, Quotient rings, First Isomorphism Theorem.

UNIT 2 (12 Hours)

Prime and Maximal Ideals, Factorization, Divisibility, units and associates, Irreducible elements, Prime elements, g.c.d., Relatively prime elements, Euclidian Domain - Definition, Examples, Existence of g.c.d., Factorization Theorem.

Unit 3 (12 Hours)

The Laplace Transforms: Definition of Laplace transforms; Transforms of elementary functions, transforming initial value problems: transforms of derivatives; Derivatives of Transforms. The Gamma functions, periodic functions. Inverse Transforms: Definition of an inverse transform, Step functions, convolution theorem.

Text Book:

- [1] N. S. Gopalakrishnan, *University Algebra*, Revised 4th edition, New Age International Publishers, 2021
- [2] Earl D Rainville and Philip E Bedient, *Elementary Differential Equations*, Pearson, 8th Ed., 2016

References:

- [1] I. N. Herstein, *Topics in Algebra*, 2nd ed, Wiley Eastern Ltd., New Delhi, 1975.

- [2] V. K. Khanna and S. K. Bhambri, *A Course in Abstract Algebra*, Vikas Publications, 1998.
- [3] M. Artin, *Algebra*, 2nd ed, Pearson, 2015.
- [4] J. A. Gallian, *Contemporary Abstract Algebra*, 10th ed, Taylor and Francis Group, 2021.
- [5] R. Murray and L. Spiegel, *Laplace Transforms*, Schaum's Series, McGraw Hill Education, 2005.
- [6] J. K. Goyal and K. P. Gupta, *Laplace and Fourier Transforms*, Pragathi Prakashan, 2016.
- [7] S. Kumar, *Integral Transform Methods in Science & Engineering*, CBS Engineering Series, CBS, 2017.

BSCMTCS502	Graph Theory	3 Credits	(36 Hours, 3 hours/week)
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Course Objectives:

1. To introduce the basic concepts of graph theory such as graphs, subgraphs, walks, paths, circuits, and trees.
2. To explore key properties of Eulerian and Hamiltonian graphs and their applications.
3. To study graph connectivity and separability, including planar graphs and their representations.
4. To understand graph colorability and the application of chromatic numbers and polynomials.
5. To apply matrix representations and vector spaces associated with graphs for deeper analysis.
6. To familiarize students with advanced topics such as digraphs, graph enumeration, and labelled trees.

Course Outcomes (COs):

By the end of the course, students will be able to:

1. Define and explain fundamental graph theory concepts like graphs, trees, and circuits.
2. Analyze and identify properties of Eulerian and Hamiltonian graphs and solve related problems.
3. Examine the connectivity and planarity of graphs and construct geometric duals.
4. Apply concepts of graph colorability and chromatic numbers to solve coloring problems.
5. Use matrix representations and understand the vector space associated with a graph.
6. Work with directed graphs and perform enumeration of graphs and counting of labelled trees.

Unit I (12 Hours)

Graph, finite and infinite graphs, Incidence and degree, Isolated vertex, Pendent vertex, Null graph, Isomorphism, Sub-graphs, Walks, Paths, Circuits, Connected and disconnected graphs, Components, Euler graphs, Operation on graphs, Hamiltonian paths and Circuits.

Unit 2 (12 Hours)

Trees and some properties of trees, Rooted and binary tree, Spanning tree and fundamental circuits. Cutsets, Properties, Fundamental cut sets, Connectivity, Separability, Planar graphs, Kuratowski's graphs, Euler's formula, Geometric duals.

Unit 3 (12 Hours)

Colourability, chromatic number, Chromatic Polynomial, Matrix representation of graphs - incident, adjacent, circuit and path matrices. Directed graphs, Matrices of digraphs.

Text Books:

- [1] Gary Chartrand and Ping Zhang, *A First Course in Graph Theory*, Dover Publications, Inc., New York, 2012.
- [2] Narsingh Deo, *Graph Theory with Applications to Engineering and Computer Science*, PHI Learning Private Limited, 2004.
- [3] R. Balakrishnan and K. Ranganathan, *A Textbook of Graph Theory*, Second Edition, Universitext, Springer, New York, 2012

Reference Books:

- [1] S. Arumugam and S. Ramachandran, *Invitation to graph theory*, Scitech Publications (India) Pvt. Ltd., 2013.
- [2] Douglas B. West, *Introduction to Graph Theory*, Pearson, 2017.
- [3] K. Chandrasekhara Rao, *Discrete Mathematics*, Narosa Publishing House, 2012.
- [4] Robin J Wilson, *Introduction to Graph Theory*, 5th Ed., Pearson, 2010.
- [5] J.A. Bondy and U.S.R. Murty, *Graph Theory with Applications*, Macmillan, 1976.

Practicals for V Semester

BSCMTPS501	Practicals on Ring Theory and Laplace Transforms	2 Credits	(4 hours/week)
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Course Objectives:

1. To learn programming skills in WxMaxima through listed programs.
2. To apply the programming skills in Science and Engineering problems.

Course Outcomes (COs):

1. Students will have the knowledge and skills to implement the programs listed below in the/Maxima programming language.
2. Students can apply these programming skills of computation in science and Engineering.

Programs:

1. Checking whether a given set is a ring with respect to given binary operations.
2. Checking whether a given set is an integral domain or field with respect to given binary operations.
3. Finding zero divisors and units in finite rings.
4. Verification of the given mapping for ring homomorphism.
5. Problems on Prime Ideals and Maximal Ideals
6. Finding the Laplace transforms of some standard functions.
7. Finding the Laplace transforms of some periodic and step functions.
8. Finding the inverse Laplace transform of simple functions
9. Verification of Convolution Theorem.
10. To solve ordinary linear differential equation using Laplace transform.

Note: The above list may be changed annually with the approval of the BOS in UG (Mathematics).

VI Semester

BSCMTCS601	Complex Analysis and Linear Algebra	3 Credits	(36 Hours, 3 hours/ week)
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Course Objectives:

1. Introduce foundational concepts in complex analysis, including algebraic properties, exponential forms, and differentiability of complex functions.
2. Develop an understanding of analytic, entire, and harmonic functions and apply them to study classical functions such as exponential, logarithmic, and trigonometric forms.
3. Familiarize students with vector space theory, emphasizing key concepts like subspaces, linear independence, and basis construction.
4. Explore the structure and properties of inner product spaces to deepen conceptual grasp and analytical reasoning in linear algebra.
5. Investigate the structure of linear transformations, their matrix representations, change of basis, and the role of eigenvalues and eigenvectors in understanding linear operators.

Course Outcomes (COs): At the end of the course, the student will be able to:

1. Analyze and interpret complex functions using Cauchy-Riemann equations and concepts of differentiability and continuity.
2. Analyze elementary complex functions and distinguish between analytic, harmonic, and entire functions in applied contexts.
3. Construct and validate vector spaces and subspaces, determining linear independence, spans, and basis dimensions.
4. Demonstrate competence in inner product spaces and apply inequalities such as Schwarz's inequality in theoretical proofs.
5. Represent linear transformations in matrix form, determine rank, nullity, and minimal polynomials, and compute eigenvalues and eigenvectors to classify linear operators.

Unit I (12 Hours)

Complex Analysis: (Recapitulation of algebra, conjugate and modulus of Complex numbers) Polar and Exponential Forms, roots of complex numbers, Functions of a Complex variable, Limits, Continuity, Differentiability, Cauchy-Riemann Equations. Analytic functions, Entire functions, and Harmonic functions.

Unit 2 (12 Hours)

Vector Spaces: Vector Space, Subspaces, Linear Span, Linear Independence, Basis and Dimension, Basis of a Vector Space, Sums of Subspaces, Quotient Space, Inner product spaces, Schwartz inequality.

Unit 3 (12 Hours)

Linear Transformations: Linearity, Rank and Nullity, Isomorphisms, Matrix Representation, minimal polynomial, Change of Basis, Space of Linear Transformations. Eigen values and Eigen vectors of Linear operators.

Text Books:

- [1] R.V. Churchill & J.W. Brown, *Complex Variables and Applications*, 8th ed., McGraw Hill 2009.
- [2] N. S Gopalakrishnan, *University Algebra*, 3rd edition, New Age International Publications, 2015.

Reference Books:

- [1] L. V. Ahlfors, *Complex Analysis*, 3rd Edition, McGraw Hill Education, 1979.
- [2] S. Ponnusamy, *Foundations of Complex Analysis*, 2nd Edition, Alpha Science International Limited, 2005.
- [3] Bruce P. Palka, *Introduction to the Theory of Function of a Complex Variable*, Springer, 1991.
- [4] Serge Lang, *Complex Analysis*, Springer, 1999.
- [5] Shanthi Narayan and P. K. Mittal, *Theory of Functions of a Complex Variable*, S. Chand Publishers, 2005.
- [6] B. S. Grewal and J. S. Grewal, *Higher Engineering Mathematics*, 42nd Edition, Khanna Publishers, New Delhi, 1996.
- [7] G. D. Birkhoff and S. MacLane, *A Brief Survey of Modern Algebra*, 2nd Edition, IBH Publishing Company, Bombay, 1967.
- [8] Stephen H. Friedberg, Arnold J. Insel, Lawrence E. Spence, *Linear Algebra*, 4th Edition, Prentice-Hall of India Pvt. Ltd., New Delhi, 2004.
- [9] Gilbert Strang, *Linear Algebra and its Applications*, Thomson, 2007.
- [10] S. Kumaresan, *Linear Algebra - A Geometric Approach*, Prentice Hall of India, 2000

BSCMTCS602	Numerical Analysis	3 Credits	(36 Hours, 3 hours/ week)
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Course Objectives:

The course will help the students

1. To introduce the fundamental concepts of Numerical Analysis including solution of Algebraic and transcendental equations and simultaneous equations.
2. To learn the deeper aspects of Polynomial interpolation
3. To find the derivatives and integrals using Numerical formulae
4. To solve the initial value problems using different methods.

Course Outcomes (COs):

At the end of this course, students will be able to:

1. Compute approximate roots of algebraic and transcendental equations using iterative methods
2. Find integrals and derivatives.
3. Solve problems on interpolation using suitable numerical techniques.
4. Obtain approximate solutions to initial value problems using various numerical techniques.

Unit I (12 Hours)

Algebraic and Transcendental Equations: Solutions to algebraic and transcendental equations - Bisection method, Regula-Falsi method, Iterative method, Newton-Raphson method.

System of Linear Algebraic Equations: Direct Methods – Gauss elimination method, Gauss-Jordan method and Iterative methods: Jacobi method, Gauss Seidal method.

Unit 2 (12 Hours)

Finite differences - Forward, Backward differences and shift operators, definitions, properties and problems. Polynomial interpolation - Newton-Gregory forward and backward interpolation formulae, Lagrange's interpolation formula. , Numerical Integration: General quadrature formula, Trapezoidal rule, Simpson's 1/3 rule, Simpson's 3/8 rule with derivations

Unit-3 (12 Hours)

Numerical Solution of Ordinary Differential Equations: Introduction, Solution by Taylor's series method, Picard's method, Euler's method, Modified Euler's method, Runge-Kutta methods, Predictor-Corrector methods - Adams-Bashforth method.

Text Books:

- [1] S. S. Sastry, Introductory methods of Numerical Analysis, 5th Edition, PHI Learning Private Limited, 2012
- [2] B .S. Grewal, Numerical methods in Engineering and Science with programs in C, C++, 9th edition, Khanna Publishers, 2010.

References:

- [1] E. Isaacson and H. B. Keller, *Analysis of Numerical Methods*, Dover Publications, 1966.
- [2] E. Kreyszig, *Advanced Engineering Mathematics*, Wiley India Pvt. Limited, 1972.
- [3] M. K. Jain, S. R. K. Iyengar, and R. K. Jain, *Numerical Methods for Scientific and Engineering Computation*, 4th Edition, New Age International, 1985.
- [4] H. C. Saxena, *Finite Difference and Numerical Analysis*, S. Chand Publishers, 2010.
- [5] B. D. Gupta, *Numerical Analysis*, Konark Publishers Pvt. Ltd., 1990

Practicals for VI Semester

BSCMTPS601	Practicals on Complex Analysis and Linear Algebra	2 Credits	(4 hours/ week)
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Course Objectives:

1. To learn programming skills in Scilab/Maxima through listed programs.
2. To apply the programming skills in Science and Engineering problems.

Course Outcomes (COs):

1. Students will have the knowledge and skills to implement the programs listed below in the Scilab/Maxima programming language.
2. Students can apply these programming skills of computation in science and Engineering.

Programs:

1. Program to find the roots of a given complex number.
2. Program to verify the Cauchy–Riemann equations in Cartesian form and test for analyticity.
3. Program to verify the Cauchy–Riemann equations in Polar form and test for analyticity.

4. Program to check whether a given function is harmonic.
5. Program to construct an analytic function using the Milne–Thomson method.
6. Program to verify linear dependence and independence.
7. Program to find basis and dimension of the subspaces.
8. Program to find the matrix of linear transformation.
9. Program to find the Eigen values and Eigen vectors of a given linear transformation.
10. Program to find the minimal polynomial of given transformation.

Note: The above list may be changed annually with the approval of the BOS in UG (Mathematics).

Compulsory Skill/Practicals
Internship/Mini Project Work
(May be offered in either IV, V or VI Semester)

BSCMTMS456	Mini project in Applications of Mathematics	2 Credits	(28 Hours, 2 hours/week)
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Internship/Mini project has to be compulsorily carried out in one of the last three semesters (i.e. Semester IV, V or VI) with a workload of 2 hours per week under the supervision of a teacher. Report on the internship/Mini project has to be submitted for evaluation.

Course Objectives:

1. To enable students to apply mathematical concepts and techniques to solve real-world problems through structured project-based learning.
2. To develop analytical and problem-solving skills by integrating mathematical modeling, computation, and logical reasoning in practical applications.

Course Outcomes (COs):

1. Students will be able to formulate and implement mathematical models to analyze and solve application-based problems in various domains.
 2. Students will demonstrate proficiency in presenting mathematical findings through structured reports and effective communication of results.
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Question Paper Patterns for Semester Exams

B.Sc. Mathematics (Three Major Scheme)

Theory (3 credit core courses)

For I/ II / III/ IV Semesters

Duration: 3 hours

Max. Marks: 80

PART -A	
I. Answer any 10 questions out of 14 questions ($10 \times 2 = 20$)	
Question Number	Unit 1 to 4
1 to 14	At least 3 questions from each unit
PART -B	
II. Answer any 3 questions from each unit ($12 \times 5 = 60$)	
Question Number	Units
1 to 5	Unit - 1
6 to 10	Unit - 2
11 to 15	Unit - 3
16 to 20	Unit -4

For V/ VI Semesters

Duration: 3 hours

Max. Marks: 80

PART -A	
I. Answer any 10 questions out of 14 questions ($10 \times 2 = 20$)	
Question Number	Unit 1 to 4
1 to 14	At least 4 questions from each unit
PART -B	
II. Answer any 12 questions out of 18 questions ($12 \times 5 = 60$)	
Question Number	Units
1 to 6	Unit - 1
7 to 12	Unit - 2
13 to 18	Unit - 3

Theory (2 credit core courses)

For III/ IV Semesters

Duration: 2 hours

Max. Marks: 40

PART -A	
I. Answer any 5 questions out of 8 questions ($5 \times 2 = 10$)	
Question Number	Unit Number
1 to 4	Unit - 1
5 to 8	Unit - 2
PART -B	
II. Answer 6 questions by choosing any three from each unit ($6 \times 5 = 30$)	
Question Number	Units
1 to 5	Unit - 1
6 to 10	Unit - 2

Practicals (2 credit courses)

For Semesters I to VI

Semester Practical Exam 40 marks + Lab Internal Assessment 10 marks = 50 marks

Semester Practical Exam:

Components	Marks
Record and Viva	10
Program (writing and execution)	30
Total	40

Lab internal assessment: Lab internal assessment marks should be based on two lab tests.

For Compulsory Skill/Practicals Course

Mini Project in Applications of Mathematics (to be carried in either IV, V or VI Semester)

Components	Marks
Project	40
Presentation	5
Viva	5
Total	50